MLE with mis-specified model

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Asymptotics when the distribution is mis-specified?

Setup: Suppose that $Y_1, ..., Y_n \overset{iid}{\sim} g$ for some (unknown) distribution with probability function g. *However*, we instead assume that $Y_i \overset{iid}{\sim} f(y|\boldsymbol{\theta})$ for some specified distribution f with parameter $\boldsymbol{\theta}$.

- Let $\boldsymbol{\theta}_0$ solve $\mathbb{E}_g\left[\frac{\partial}{\partial\boldsymbol{\theta}}\log f(Y|\boldsymbol{\theta})\right] = \mathbf{0}$.
- Let $\widehat{\boldsymbol{\theta}}$ solve $\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}}\log f(Y_i|\boldsymbol{\theta}) = \mathbf{0}$

Key results: Under regularity conditions,

sandwich variance

- $\widehat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$
- $\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S}(\boldsymbol{\theta}_0))$ where

bread

$$\mathbf{S}(\boldsymbol{\theta}_0) = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$$

meat (cheese)

$$\mathbf{A} = -\mathbb{E}_g\left[\frac{\partial^2}{\partial\boldsymbol{\theta}_0^2}\log f(Y_i|\boldsymbol{\theta}_0)\right] \qquad \mathbf{B} = Var_g\left(\frac{\partial}{\partial\boldsymbol{\theta}_0}\log f(Y_i|\boldsymbol{\theta}_0)\right)$$

if f is correct, then $A = B = \mathcal{I}_1(\theta_0) \Rightarrow S(\theta_0) = \mathcal{I}_1^{-1}(\theta_0)$

Example: linear regression

$$\mathrm{Var}(Y) = \mathbb{E}[(Y-\mu)(Y-\mu)^T]$$
$$= \mathbb{E}[YY^T] - \mu\mu^T$$
$$\mu = \mathbb{E}[Y]$$

Assumed model: $Y_i|\mathbf{x}_i \sim N(\mathbf{x}_i^T\beta, \sigma^2)$

$\hat{\beta}$ solves $\sum_{i=1}^{n} (Y_i - x_i^T\beta)x_i = 0$

β^0 solves $\mathbb{E}_g[(Y_i - x_i^T\beta)x_i] = 0$

$$S(\beta^0) = A^{-1}BA^{-1}$$

$$B = \mathrm{Var}_g((Y_i - x_i^T\beta^0)x_i)$$

$$= \mathbb{E}_g[(Y_i - x_i^T\beta^0)x_i x_i^T(Y_i - x_i^T\beta^0)] - \underbrace{\mathbb{E}_g[(Y_i - x_i^T\beta^0)x_i]}_{=0}\mathbb{E}_g[(Y_i - x_i^T\beta^0)x_i]^T$$

$$= \mathbb{E}_g[(Y_i - x_i^T\beta^0)^2 x_i x_i^T]$$

$$A = -\mathbb{E}_g\left[\frac{\partial}{\partial\beta^0}(Y_i - x_i^T\beta^0)x_i\right]$$

$$= -\mathbb{E}_g[-x_i x_i^T] = \mathbb{E}_g[x_i x_i^T]$$

Example: linear regression

compare to "naive" estimator:

$\hat{Var}(\hat{\beta}) = \hat{\sigma}^2 (X^T X)^{-1}$

(correct <u>if</u> model is correct)

Assumed model: $Y_i | \mathbf{x}_i \sim N(\mathbf{x}_i^T \beta, \sigma^2)$

$$S(\beta^0) = A^{-1} B A^{-1}$$

$$B = \mathbb{E}_g \left[(Y_i - x_i^T \beta^0)^2 x_i x_i^T \right]$$

<u>Plug-in estimator</u>: $\hat{B} = \frac{1}{n} \sum_{i=1}^{n} (Y_i - x_i^T \hat{\beta})^2 x_i x_i^T$

$$= \frac{1}{n} X^T W X$$

$$W = \mathrm{diag}\left((Y_i - \hat{Y}_i)^2 \right)$$

$$A = \mathbb{E}_g \left[x_i x_i^T \right]$$

<u>Plug-in estimator</u>: $\hat{A} = \frac{1}{n} \sum_{i=1}^{n} x_i x_i^T = \frac{1}{n} X^T X$

$$\Rightarrow \hat{S} = \hat{A}^{-1} \hat{B} \hat{A}^{-1} = \left(\frac{1}{n} X^T X\right)^{-1} \left(\frac{1}{n} X^T W X\right) \left(\frac{1}{n} X^T X\right)^{-1}$$

$$= n (X^T X)^{-1} X^T W X (X^T X)^{-1}$$

$$\sqrt{n}(\hat{\beta} - \beta^0) \approx N(0, \hat{S})$$

$$\Rightarrow Var(\hat{\beta}) \approx (X^T X)^{-1} X^T W X (X^T X)^{-1}$$

(<u>robust</u> estimator)